

How does cosmological expansion affect black hole masses and rotation?

Cosmological accretion (matter accumulation or growth) is a fundamental mechanism that profoundly influences the masses and spin properties of black holes on cosmological timescales. Initially low-mass and slowly rotating primordial black holes (PBHs) can evolve through this process into much more massive and rapidly rotating objects.

Effects on Mass:

General Mass Increase: Cosmological expansion causes the masses of primordial black holes to increase significantly over time, shifting the mass distribution toward higher masses and creating a high-mass tail in the population.

Changes in the Mass Ratio in Binary Systems: When accretion occurs in a binary black hole system, the component with the lower mass accretes matter at a proportionally higher rate. As a direct result, the mass ratio ($q = m_2/m_1$) in the binary system tends to increase over time; that is, as time progresses, the masses of the two black holes converge more closely.

Effects on Spin:

From Initial Spin to Acceleration: Since primordial black holes formed during the early, radiation-dominated phase of the universe, their initial spin rates are expected to be very low (close to zero). However, accretion is a very powerful mechanism that allows these black holes to reach enormous spin rates.

Maximum Rotation Speed and Mass-Rotation Correlation: This growth process, which occurs via the accretion disk, can increase the black hole's rotation up to a value of $X_{\text{max}} = 0.998$ a dimensionless parameter—due to radiation losses. As a result of this process, a positive correlation arises between mass and spin rate in black holes. In other words, black holes whose masses increase more significantly through cosmological growth also have higher spin rates.

In particular, black hole mergers such as GW231123—which feature the most massive and surprisingly rapidly rotating (high-spin) components ever detected in the universe—can be explained naturally and consistently by the process through which small, primordial black holes undergo this efficient cosmological growth mechanism.